



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

School of Computer Engineering

Manipal – 576 104

Operating Systems Lab [CSS-2211]

Fourth Semester

B. Tech. (CSE Stream)



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that Ms. Hridini Pandey

Reg. No.: 240911482

Section: ITB

Roll No.: 39

has satisfactorily completed the lab exercises prescribed for
OPERATING SYSTEMS LAB [CSS 2211] of Second Year
B.Tech. Degree at MIT, Manipal, in the academic year
2025-2026.

Date:

Signature

Faculty in Charge

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LAB NO: 1

UNIX SHELL COMMANDS

Lab Exercises:

1. Create a folder with your <name >_registration number in the /home folder. Use this folder to save rest of lab exercises. For each lab create a separate folder ex: Lab1, Lab2... Lab11.

```
4ITB2@debian:~$ cd hridini_240911482
4ITB2@debian:~/hridini_240911482$ mkdir lab1
4ITB2@debian:~/hridini_240911482$ cd lab1
4ITB2@debian:~/hridini_240911482/lab1$ cat sample.txt
cat: sample.txt: No such file or directory
4ITB2@debian:~/hridini_240911482/lab1$ touch sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ cat sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ head sample.txt
hi
this is os lab
testing...
blah blah
abc
xyz
ong
fr
4ITB2@debian:~/hridini_240911482/lab1$ mkdir f1
4ITB2@debian:~/hridini_240911482/lab1$ cp sample.txt/f1
cp: missing destination file operand after 'sample.txt/f1'
Try 'cp --help' for more information.
4ITB2@debian:~/hridini_240911482/lab1$ cd..
bash: cd.: command not found
4ITB2@debian:~/hridini_240911482/lab1$ mkdir f1
```

2. Execute and write output of any 5 important commands explained so far in this manual. Save the output in Lab1_1.txt . Commands and the output I copied from the command prompt.

```
lab1_1.txt
~/hridini_240911482/lab1
Open Save

1 4ITB2@debian:~/lab1/hridini_240911482$ cd lab1
2 4ITB2@debian:~/lab1/hridini_240911482/lab1$ pwd
3 /home/4ITB2/lab1/hridini_240911482/lab1
4 4ITB2@debian:~/lab1/hridini_240911482/lab1$ date
5 Wednesday 14 January 2026 09:34:34 AM IST
6
7 4ITB2@debian:~/hridini_240911482/lab1$ head -n 5 sample.txt
8 hi
9 this is os lab
10 testing...
11 blah blah
12 abc
13 4ITB2@debian:~/hridini_240911482/lab1$ cp sample.txt f1
14 4ITB2@debian:~/hridini_240911482/lab1$ cd f1
15 4ITB2@debian:~/hridini_240911482/lab1/f1$ tail sample.txt
16 hi
17 this is os lab
18 testing...
19 blah blah
20 abc
21 xyz
22 ong
23 fr
24
25 4ITB2@debian:~/hridini_240911482/lab1$ ls
26 f1 sample.txt
27 4ITB2@debian:~/hridini_240911482/lab1$ cd ..
28 4ITB2@debian:~/hridini_240911482$ ls
29 f1 lab1
```

3. Explore the following commands along with their various options. (Some of the options are specified in the bracket)

- cat (variation used to create a new file and append to existing file)
- head and tail (-n, -c)
- cp (-n, -i, -f)
- mv (-f, -i) [try (i) mv dir1 dir2 (ii) mv file1 file2 file3 ... directory]
- rm (-r, -i, -f)
- rmdir (-r, -f)
- find (-name, -type)

Save the output in Lab1_2.txt, Commands and the output I copied from the command prompt.

3. List all the file names satisfying following criteria

a. has the extension .txt.

```
4ITB2@debian: ~/hridini_240911482
File Edit View Search Terminal Help
mkdir: cannot create directory 'f1': File exists
4ITB2@debian:~/hridini_240911482/lab1$ mkdir f1
4ITB2@debian:~/hridini_240911482/lab1$ cd ..
4ITB2@debian:~/hridini_240911482$ mkdir f1
4ITB2@debian:~/hridini_240911482$ cd lab1
4ITB2@debian:~/hridini_240911482/lab1$ cp sample.txt/f1
cp: missing destination file operand after 'sample.txt/f1'
Try 'cp --help' for more information.
4ITB2@debian:~/hridini_240911482/lab1$ head -n 5 sample.txt
hi
this is os lab
testing...
blah blah
abc
4ITB2@debian:~/hridini_240911482/lab1$ cp sample.txt f1
4ITB2@debian:~/hridini_240911482/lab1$ cd f1
4ITB2@debian:~/hridini_240911482/lab1/f1$ tail sample.txt
hi
this is os lab
testing...
blah blah
abc
xyz
ong
fr
4ITB2@debian:~/hridini_240911482/lab1/f1$ cd ..
4ITB2@debian:~/hridini_240911482/lab1$ ls
f1 sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ cd ..
4ITB2@debian:~/hridini_240911482$ ls
f1 lab1
4ITB2@debian:~/hridini_240911482$ cd lab1
4ITB2@debian:~/hridini_240911482/lab1$ cat sample.txt
hi
this is os lab
testing...
blah blah
abc
xyz
ong
fr
4ITB2@debian:~/hridini_240911482/lab1$ touch lab1_1.txt
4ITB2@debian:~/hridini_240911482/lab1$ ^C
4ITB2@debian:~/hridini_240911482/lab1$ tail -c 4 sample.txt
fr
```

b. containing atleast one digit.

c. having minimum length of

4.

d. does not contain any of the vowels as the start letter.



4ITB2@debian: ~/hridini_240911482

File Edit View Search Terminal Help

```
4ITB2@debian:~/hridini_240911482/lab1$ tail -n 5 sample.txt
```

blah blah

abc

xyz

ong

fr

```
4ITB2@debian:~/hridini_240911482/lab1$ tail sample.txt
```

hi

this is os lab

testing...

blah blah

abc

xyz

ong

fr

```
4ITB2@debian:~/hridini_240911482/lab1$ tail -i 5 sample.txt
```

tail: invalid option -- 'i'

Try 'tail --help' for more information.

```
4ITB2@debian:~/hridini_240911482/lab1$ find sample.txt
```

sample.txt

```
4ITB2@debian:~/hridini_240911482/lab1$ cd f1
```

```
4ITB2@debian:~/hridini_240911482/lab1/f1$ touch ex.txt
```

```
4ITB2@debian:~/hridini_240911482/lab1/f1$ cd ..
```

```
4ITB2@debian:~/hridini_240911482/lab1$ find ex.txt
```

find: 'ex.txt': No such file or directory

```
4ITB2@debian:~/hridini_240911482/lab1$ find .-name*.ex
```

find: './-name*.ex': No such file or directory

```
4ITB2@debian:~/hridini_240911482/lab1$ find . -name "ex.txt"
```

./f1/ex.txt

```
4ITB2@debian:~/hridini_240911482/lab1$ cd -i sample.txt s2.txt
```

bash: cd: -i: invalid option

cd: usage: cd [-L|[-P [-e]] [-@]] [dir]

```
4ITB2@debian:~/hridini_240911482/lab1$ cd -i sample.txt s2.txt
```

bash: cd: -i: invalid option

cd: usage: cd [-L|[-P [-e]] [-@]] [dir]

```
4ITB2@debian:~/hridini_240911482/lab1$ cp -i sample.txt s2.txt
```

```
4ITB2@debian:~/hridini_240911482/lab1$ cp -i sample.txt s2.txt
```

cp: overwrite 's2.txt'? n

```
4ITB2@debian:~/hridini_240911482/lab1$ cp -n sample.txt s2.txt
```

```
4ITB2@debian:~/hridini_240911482/lab1$ cp -n sample.txt s2.txt
```

```
4ITB2@debian:~/hridini_240911482/lab1$ cp -n lab1_1.txt s2.txt
```

```
4ITB2@debian:~/hridini_240911482/lab1$ cp -f lab1_1.txt s2.txt
```

```
4ITB2@debian:~/hridini_240911482/lab1$ mv sample.txt f1
```

```
4ITB2@debian:~/hridini_240911482/lab1$ cd f1
```

```
4ITB2@debian:~/hridini_240911482/lab1/f1$ ls
```

ex.txt sample.txt

```
4ITB2@debian: ~/hridini_240911482
File Edit View Search Terminal Help

4ITB2@debian:~/hridini_240911482/lab1$ cd f1
4ITB2@debian:~/hridini_240911482/lab1/f1$ ls
ex.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1/f1$ cd ..
4ITB2@debian:~/hridini_240911482/lab1$ ls
f1 lab1_1.txt s2.txt
4ITB2@debian:~/hridini_240911482/lab1$ rm -i s2.txt
rm: remove regular file 's2.txt'? n
4ITB2@debian:~/hridini_240911482/lab1$ cd f1
4ITB2@debian:~/hridini_240911482/lab1/f1$ mv sample.txt ..
4ITB2@debian:~/hridini_240911482/lab1/f1$ cd ..
4ITB2@debian:~/hridini_240911482/lab1$ ls
f1 lab1_1.txt s2.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ rm -r f1
4ITB2@debian:~/hridini_240911482/lab1$ ls
lab1_1.txt s2.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ ls *.txt
lab1_1.txt s2.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ touch this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls *[0-9]*
lab1_1.txt s2.txt this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls ????*
lab1_1.txt s2.txt sample.txt this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls ??????*
lab1_1.txt sample.txt this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls [!aeiouAEIOU]*
lab1_1.txt s2.txt sample.txt this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls !*.txt
ls [!aeiouAEIOU]*.txt
lab1_1.txt s2.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ ls [!aeiouAEIOU]*.txt
lab1_1.txt s2.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ touch abc.txt
4ITB2@debian:~/hridini_240911482/lab1$ ls
abc.txt lab1_1.txt s2.txt sample.txt this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls [!aeiouAEIOU]*.txt
lab1_1.txt s2.txt sample.txt
4ITB2@debian:~/hridini_240911482/lab1$ ls !(*.txt)
this123.c
4ITB2@debian:~/hridini_240911482/lab1$ ls ??????
s2.txt
4ITB2@debian:~/hridini_240911482/lab1$ cd ..
4ITB2@debian:~/hridini_240911482$ cd lab1
4ITB2@debian:~/hridini_240911482/lab1$ touch lab1.txt
4ITB2@debian:~/hridini_240911482/lab1$ cd ..
4ITB2@debian:~/hridini_240911482$
```


LAB NO: 2

ADVANCED UNIX SHELL COMMANDS

Lab Exercises

1. Execute all the commands explained in this section and write the output.

```
4ITB2@debian:~/hridini_240911482/lab2$ touch fruitlist.txt
4ITB2@debian:~/hridini_240911482/lab2$ grep apple fruitlist.txt
apple
pineapple
fruit-apple
4ITB2@debian:~/hridini_240911482/lab2$ wc fruitlist.txt
11 10 73 fruitlist.txt
4ITB2@debian:~/hridini_240911482/lab2$ sort fruitlist.txt

apple
banana
fruit-apple
grape
mango
orange
peach
pear
pineapple
tomato
```

```
4ITB2@debian:~/hridini_240911482/lab2$ grep -v apple fruitlist.txt
```

```
grape  
orange  
banana  
pear  
peach  
mango  
tomato
```

```
4ITB2@debian:~/hridini_240911482/lab2$ grep -x apple fruitslist.txt  
grep: fruitslist.txt: No such file or directory
```

```
4ITB2@debian:~/hridini_240911482/lab2$ grep -x apple fruitlist.txt  
apple
```

```
4ITB2@debian:~/hridini_240911482/lab2$ grep ^p fruitlist.txt  
pineapple  
pear  
peach
```

```
4ITB2@debian:~/hridini_240911482/lab2$ sort -r fruitlist.txt
```

```
tomato  
pineapple  
pear  
peach  
orange  
mango  
grape  
fruit-apple  
banana  
apple
```

```
4ITB2@debian:~/hridini_240911482/lab2$ cut -c1-3 fruitlist.txt
```

```
app  
gra  
ora  
pin  
fru  
ban  
pea  
pea  
man  
tom
```

```
4ITB2@debian:~/hridini_240911482/lab2$ ls -l | sed -n '/^d/p'
```

```
4ITB2@debian:~/hridini_240911482/lab2$ ps
```

PID	TTY	TIME	CMD
7846	pts/0	00:00:00	bash
408038	pts/0	00:00:00	ps

```
4ITB2@debian:~/hridini_240911482/lab2$ tr 'a-z' 'A-Z' < fruitlist.txt
```

```
ps
```

```
APPLE
```

```
GRAPE
```

```
ORANGE
```

```
PINEAPPLE
```

```
FRUIT-APPLE
```

```
BANANA
```

```
PEAR
```

```
PEACH
```

```
MANGO
```

```
TOMATO
```

```
TOMATO
```

PID	TTY	TIME	CMD
7846	pts/0	00:00:00	bash
408040	pts/0	00:00:00	ps

```
4ITB2@debian:~/hridini_240911482/lab2$ echo "2^5" | bc
```

```
32
```

```
4ITB2@debian:~/hridini_240911482/lab2$ sort fruitlist.txt
```

```
apple
```

```
banana
```

```
fruit-apple
```

```
grape
```

```
mango
```

```
orange
```

```
peach
```

```
pear
```

```
pineapple
```

```
tomato
```

2. Write grep commands to do the following activities:

- To select the lines from a file that have exactly two characters.
- To select the lines from a file that start with the upper case letter.
- To select the lines from a file that end with a period.
- To select the lines in a file that has one or more blank spaces.
- To select the lines in a file and direct them to another file which has digits as one of the characters in that line.

```
4ITB2@debian:~/hridini_240911482/lab2$ touch data.txt
4ITB2@debian:~/hridini_240911482/lab2$ grep -E '^..$' data.txt
4ITB2@debian:~/hridini_240911482/lab2$ grep -E '^..$' data.txt
ch
me
un
4ITB2@debian:~/hridini_240911482/lab2$ grep -E '^[A-Z]' data.txt
Apple
ALL CAPS
4ITB2@debian:~/hridini_240911482/lab2$ grep -E '\.$' data.txt
aur lines lo.
pineapple.
4ITB2@debian:~/hridini_240911482/lab2$ grep -E ' +' data.txt
yeh kaun hai
ALL CAPS
aur lines lo.
4ITB2@debian:~/hridini_240911482/lab2$ grep -E '[0-9]' data.txt > digits.txt
```

3. Create file studentInformation.txt using vi editor which contains details in the following format.

RegistrationNo:Name:Department:Branch:Section:Sub1:Sub2:Sub3
1234:XYZ:ICT:CCE:A:80:60:70 ... (add atleast 10 rows)

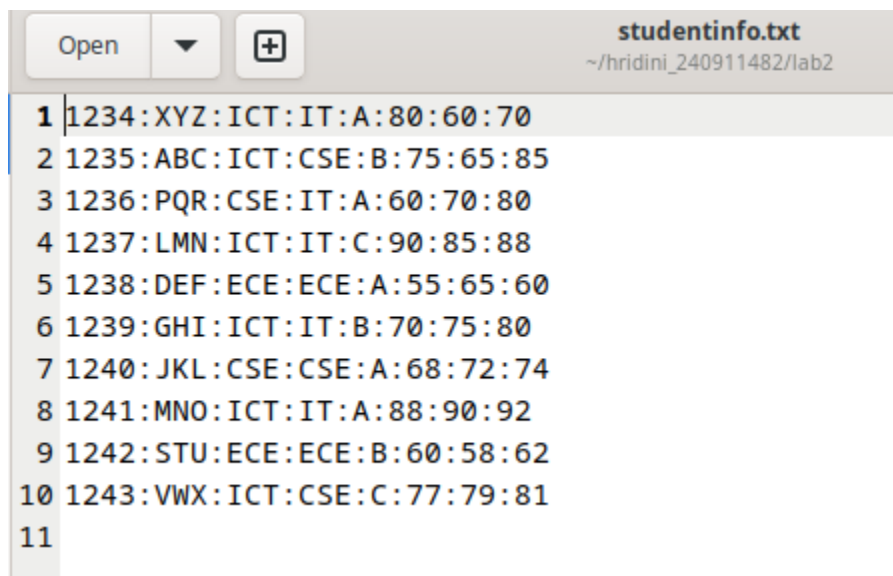
i) Display the number students(only count) belonging to ICT department.

ii) Replace all occurrences of IT branch with “Information Technology” and save

the output to ITStudents.txt

iii) Display the average marks of student with the given registration number

“1234” (or any specific existing number).

A screenshot of a text editor window titled 'studentinfo.txt' with a path of '~/.hridini_240911482/lab2'. The editor shows a list of 10 student records, each on a new line, numbered 1 to 10. Each record follows the format: RegistrationNo:Name:Department:Branch:Section:Sub1:Sub2:Sub3. The records are: 1 1234:XYZ:ICT:IT:A:80:60:70, 2 1235:ABC:ICT:CSE:B:75:65:85, 3 1236:PQR:CSE:IT:A:60:70:80, 4 1237:LMN:ICT:IT:C:90:85:88, 5 1238:DEF:ECE:ECE:A:55:65:60, 6 1239:GHI:ICT:IT:B:70:75:80, 7 1240:JKL:CSE:CSE:A:68:72:74, 8 1241:MNO:ICT:IT:A:88:90:92, 9 1242:STU:ECE:ECE:B:60:58:62, 10 1243:VWX:ICT:CSE:C:77:79:81. Line 11 is empty.

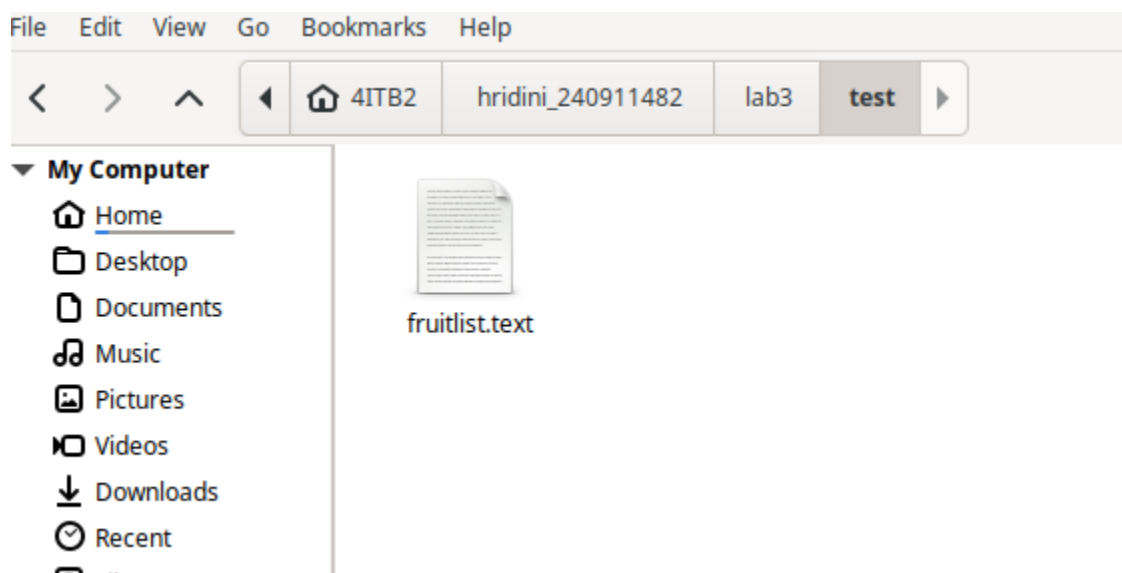
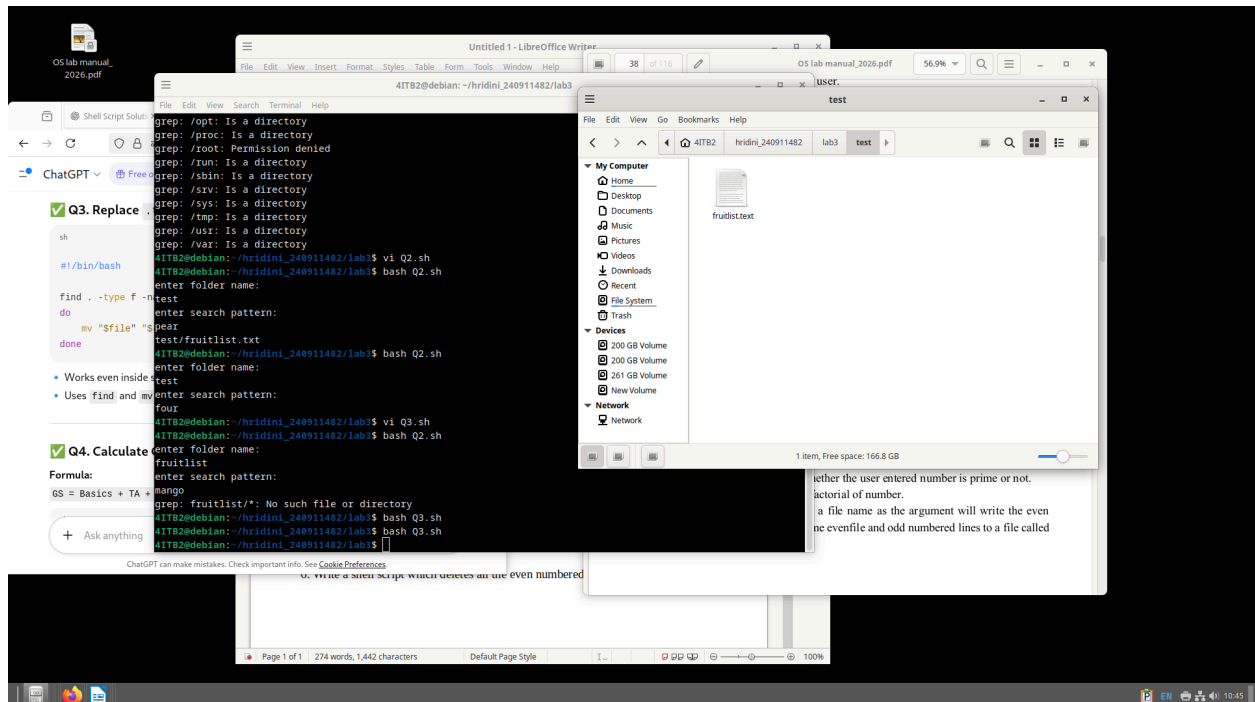
```
4ITB2@debian:~/hridini_240911482/lab2$ grep -E '[0-9]' data.txt > digits.txt
4ITB2@debian:~/hridini_240911482/lab2$ vi studentinfo.txt
4ITB2@debian:~/hridini_240911482/lab2$ grep -c ':ICT:' studentInformation.txt
grep: studentInformation.txt: No such file or directory
4ITB2@debian:~/hridini_240911482/lab2$ sed 's/:IT/:Information Technology:/g' studentinfo.txt > ITStudents.txt
4ITB2@debian:~/hridini_240911482/lab2$ grep '^1234:' studentinfo.txt | \
awk -F':' '{ avg=($6+$7+$8)/3; print avg }'
70
4ITB2@debian:~/hridini_240911482/lab2$
```

LAB NO: 3

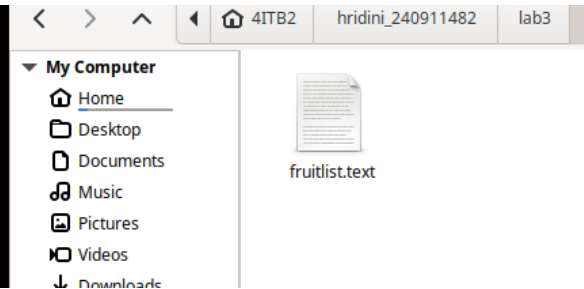
UNIX SHELL PROGRAMMING (SHELL SCRIPTING)

Lab Exercises

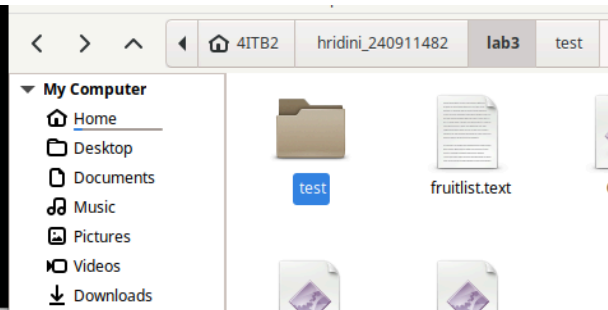
1. Write a shell script to find whether a given file is the directory or regular file.



```
4ITB2@debian:~/hridini_240911482/lab3$ vi Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
fruitlist
enter search pattern:
mango
grep: fruitlist/*: No such file or directory
4ITB2@debian:~/hridini_240911482/lab3$ bash Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$
```



```
four
4ITB2@debian:~/hridini_240911482/lab3$ vi Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
fruitlist
enter search pattern:
mango
grep: fruitlist/*: No such file or directory
4ITB2@debian:~/hridini_240911482/lab3$ bash Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$
```



fruitlist.text
~/hridini_240911482/lab3

```
1 apple
2 orange
3 fruit-apple
4 pear
5 mango
6 Example: huh this is not an example, this is an ex. here
```

Open + Save ~/hridini_240911482/lab3/test

fruitlist.txt x fruitlist.txt x

```
1 apple
2 grape
3 orange
4 pineapple
5 fruit-apple
6 banana
7 pear
8 peach
9 mango
10 tomato
11
```



```
4ITB2@debian:~$ cd hridini_240911482
4ITB2@debian:~/hridini_240911482$ mkdir lab3
4ITB2@debian:~/hridini_240911482$ cd lab3
4ITB2@debian:~/hridini_240911482/lab3$ #!/bin/sh
4ITB2@debian:~/hridini_240911482/lab3$ vi script.sh
4ITB2@debian:~/hridini_240911482/lab3$ ./script.sh
bash: ./script.sh: Permission denied
4ITB2@debian:~/hridini_240911482/lab3$ source script.sh
HRIDINI
4ITB2@debian:~/hridini_240911482/lab3$ echo $value
HRIDINI
4ITB2@debian:~/hridini_240911482/lab3$ result=$(./script.sh
bash: ./script.sh: Permission denied
4ITB2@debian:~/hridini_240911482/lab3$ chmod +x script.sh
4ITB2@debian:~/hridini_240911482/lab3$ result=$(./script.sh
4ITB2@debian:~/hridini_240911482/lab3$ echo $result
HRIDINI
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash script.sh
HRIDINI
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
script.sh
Q1.sh: line 5: syntax error near unexpected token `then'
Q1.sh: line 5: `if[-d "$filename"]; then'
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
script.sh
Q1.sh: line 5: syntax error near unexpected token `then'
Q1.sh: line 5: `if[-d "$filename" ]; then'
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
script.sh
```

2. Write a shell script to list all files (only file names) containing the input pattern

(string) in the folder entered by the user.

3. Write a shell script to replace all files with .txt extension with .text in the current directory. This has to be done recursively i.e if the current folder contains a folder

“OS” with abc.txt then it has to be changed to abc.text (Hint: use find, mv)

4. Write a shell script to calculate the gross salary. $GS = \text{Basics} + TA + 10\% \text{ of Basics}$.

Floating point calculations has to be performed.

5. Write a program to copy all the files (having file extension input by the user) in the

current folder to the new folder input by the user. ex: user enter .text TEXT then all

files with .text should be moved to TEXT folder. This should be done only at single

level. i.e if the current folder contains a folder name ABC which has .txt files then

these files should not be copied to TEXT.

Write a shell script to modify all occurrences of “ex:” with

“Example:” in all the files present

31LAB NO: 3

in current folder only if “ex:” occurs at the start of the line or after a period (.). Example:

if a file contains a line: “ex: this is first occurrence so should be replaced” and “second

ex: should not be replaced as it occurs in the middle of the sentence.”6. Write a shell script which deletes all the even

numbered lines in a text file.

```
script.sh
Q1.sh: line 5: syntax error near unexpected token `then'
Q1.sh: line 5: `if[-d "$filename" ]; then'
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
script.sh
Q1.sh: line 5: [-d: command not found
Q1.sh: line 7: [-f: command not found
script.sh does not exist
4ITB2@debian:~/hridini_240911482/lab3$ vi Q1.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
script.sh
script.sh is regular file
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
lab2
lab2 does not exist
4ITB2@debian:~/hridini_240911482/lab3$ mkdir test
4ITB2@debian:~/hridini_240911482/lab3$ bash Q1.sh
enter file or directory name:
test
test is a directory
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
bash: Q2.sh: No such file or directory
4ITB2@debian:~/hridini_240911482/lab3$ vi Q2.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q2.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
test
enter search pattern:
.txt
grep: test: Is a directory
grep: /bin: Is a directory
grep: /boot: Is a directory
grep: /dev: Is a directory
grep: /etc: Is a directory
grep: /home: Is a directory
/initrd.img
/initrd.img old
```

```
grep: /sys: Is a directory
grep: /tmp: Is a directory
grep: /usr: Is a directory
grep: /var: Is a directory
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
test
enter search pattern:
fruit
grep: test: Is a directory
grep: /bin: Is a directory
grep: /boot: Is a directory
grep: /dev: Is a directory
grep: /etc: Is a directory
grep: /home: Is a directory
grep: /lib: Is a directory
grep: /lib32: Is a directory
grep: /lib64: Is a directory
grep: /libx32: Is a directory
grep: /lost+found: Permission denied
grep: /media: Is a directory
grep: /mnt: Is a directory
grep: /opt: Is a directory
grep: /proc: Is a directory
grep: /root: Permission denied
grep: /run: Is a directory
grep: /sbin: Is a directory
grep: /srv: Is a directory
grep: /sys: Is a directory
grep: /tmp: Is a directory
grep: /usr: Is a directory
grep: /var: Is a directory
4ITB2@debian:~/hridini_240911482/lab3$ vi Q2.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
test
enter search pattern:
pear
test/fruitlist.txt
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
test
enter search pattern:
four
4ITB2@debian:~/hridini_240911482/lab3$ vi Q2.sh
```

```
enter folder name:
test
enter search pattern:
four
4ITB2@debian:~/hridini_240911482/lab3$ vi Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q2.sh
enter folder name:
fruitlist
enter search pattern:
mango
grep: fruitlist/*: No such file or directory
4ITB2@debian:~/hridini_240911482/lab3$ bash Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q3.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
Q4.sh: line 3: syntax error near unexpected token `hit'
Q4.sh: line 3: `$bc(hit enter)5+2'
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
Q4.sh: line 3: syntax error near unexpected token `hit'
Q4.sh: line 3: `$bc(hit enter)5+2'
4ITB2@debian:~/hridini_240911482/lab3$ vi Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
7
4ITB2@debian:~/hridini_240911482/lab3$ vi Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
7
4ITB2@debian:~/hridini_240911482/lab3$ vi Q4.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q4.sh
Enter Basic Salary:
5000
Enter TA:
600
Gross Salary is 6100.00
4ITB2@debian:~/hridini_240911482/lab3$ vi Q5.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q5.sh
Enter file extension (example: .text):
.text
Enter destination folder name:
test
Files copied successfully
4ITB2@debian:~/hridini_240911482/lab3$ bash Q5.sh
Enter file extension (example: .text):
.sh
```

```
4ITB2@debian:~/hridini_240911482/lab3$ bash Q5.sh
Enter file extension (example: .text):
.sh
Enter destination folder name:
test
Files copied successfully
4ITB2@debian:~/hridini_240911482/lab3$ bash Q5b.sh
bash: Q5b.sh: No such file or directory
4ITB2@debian:~/hridini_240911482/lab3$ vi Q5b.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q5b.sh
4ITB2@debian:~/hridini_240911482/lab3$ vi Q6.sh
4ITB2@debian:~/hridini_240911482/lab3$ bash Q6.sh
Enter file name:
fruitlist
sed: can't read fruitlist: No such file or directory
4ITB2@debian:~/hridini_240911482/lab3$ bash Q6.sh
Enter file name:
fruitlist.text
```

LAB NO: 4

PROCESS AND THREAD MANAGEMENT

Lab Exercises

1. Write a C program to create a child process. Display different messages in parent process and child process. Display PID and PPID of both parent and child process.

Block parent process until child completes using wait system call.

2. Write a C program to load the binary executable of the previous program in a child process using exec system call.

3. Create a zombie (defunct) child process (a child with exit() call, but no corresponding wait() in the sleeping parent) and allow the init process to adopt it (after parent terminates). Run the process as background process and run "ps" Command.

```

ex1.c ✕
1  #include <stdio.h>
2  #include <unistd.h>
3  #include <sys/wait.h>
4  #include <stdlib.h>
5
6  int main() {
7      pid_t pid = fork();
8
9      if (pid < 0) {
10         perror("fork failed");
11         exit(1);
12     }
13     else if (pid == 0) {    // Child
14         printf("Child Process\n");
15         printf("PID = %d\n", getpid());
16         printf("PPID = %d\n", getppid());
17         exit(0);
18     }
19     else {                // Parent
20         wait(NULL);
21         printf("Parent Process\n");
22         printf("PID = %d\n", getpid());
23         printf("PPID = %d\n", getppid());
24     }
25     return 0;
26 }

```

```

ex2.c ✕
1  #include <stdio.h>
2  #include <unistd.h>
3  #include <sys/wait.h>
4  #include <stdlib.h>
5
6  int main() {
7      pid_t pid = fork();
8
9      if (pid == 0) {
10         // Child process
11         execl("./ex1", "ex1", NULL);
12
13         // Runs only if exec fails
14         perror("exec failed");
15         exit(1);
16     }
17     else {
18         // Parent process
19         wait(NULL);
20         printf("Parent: ex1 execution completed\n");
21     }
22
23     return 0;
24 }

```



```
ex3.c ✕
1  #include <stdio.h>
2  #include <unistd.h>
3  #include <stdlib.h>
4
5  int main() {
6      pid_t pid = fork();
7
8      if (pid == 0) {    // Child
9          printf("Child exiting\n");
10         exit(0);
11     }
12     else {             // Parent
13         sleep(20);     // Parent sleeps, no wait()
14         printf("Parent exiting\n");
15     }
16     return 0;
17 }
```

```
4ITB2@debian:~/hridini_240911482/week4$ gcc example2.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
usage: ./a.out command [args...]
4ITB2@debian:~/hridini_240911482/week4$ vi ex1.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex1.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Child Process
PID = 379259
PPID = 379258
Parent Process
PID = 379258
PPID = 17658
4ITB2@debian:~/hridini_240911482/week4$ gcc ex1.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Child Process
PID = 379269
PPID = 379268
Parent Process
PID = 379268
PPID = 17658
4ITB2@debian:~/hridini_240911482/week4$ gcc ex1.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Child Process
PID = 380564
PPID = 380563
Parent Process
PID = 380563
PPID = 17658
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Child Process
PID = 381854
PPID = 381853
Parent Process
PID = 381853
PPID = 17658
4ITB2@debian:~/hridini_240911482/week4$ vi ex2.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex2.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
exec failed: No such file or directory
```

```

4ITB2@debian:~/hridini_240911482/week4$ ls
a.out ex1.c ex2.c example2.c example.c fork1.c hello.c wait1.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex1.c -o ex1
4ITB2@debian:~/hridini_240911482/week4$ ls
a.out ex1 ex1.c ex2.c example2.c example.c fork1.c hello.c wait1.c
4ITB2@debian:~/hridini_240911482/week4$ vi ex2.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex2.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Child Process
PID = 403793
PPID = 403792
Parent Process
PID = 403792
PPID = 403791
Parent: ex1 execution completed
4ITB2@debian:~/hridini_240911482/week4$ vi ex3.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex2.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out &
ps
[1] 419271
Child Process
PID = 419274
PPID = 419273
Parent Process
PID = 419273
PPID = 419271
Parent: ex1 execution completed
    PID TTY          TIME CMD
    17658 pts/0        00:00:00 bash
    419272 pts/0        00:00:00 ps
[1]+  Done                  ./a.out
4ITB2@debian:~/hridini_240911482/week4$ ./a.out &
[1] 419275
4ITB2@debian:~/hridini_240911482/week4$ Child Process
PID = 419277
PPID = 419276
Parent Process
PID = 419276
PPID = 419275

```

```

4ITB2@debian:~/hridini_240911482/week4$ Child Process
PID = 419277
PPID = 419276
Parent Process
PID = 419276
PPID = 419275
Parent: ex1 execution completed
ps
  PID TTY          TIME CMD
 17658 pts/0        00:00:00 bash
 419278 pts/0        00:00:00 ps
[1]+  Done                  ./a.out
4ITB2@debian:~/hridini_240911482/week4$ vi ex4.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex4.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex3.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out & ps
[1] 425741
Child exiting
  PID TTY          TIME CMD
 17658 pts/0        00:00:00 bash
 425741 pts/0        00:00:00 a.out
 425742 pts/0        00:00:00 ps
 425743 pts/0        00:00:00 a.out
4ITB2@debian:~/hridini_240911482/week4$ ./a.out &
[2] 427032
4ITB2@debian:~/hridini_240911482/week4$ Child exiting
ps
  PID TTY          TIME CMD
 17658 pts/0        00:00:00 bash
 425741 pts/0        00:00:00 a.out
 425743 pts/0        00:00:00 a.out
 427032 pts/0        00:00:00 a.out
 427033 pts/0        00:00:00 a.out
 427034 pts/0        00:00:00 ps
4ITB2@debian:~/hridini_240911482/week4$ Parent exiting

[1]-  Done                  ./a.out
4ITB2@debian:~/hridini_240911482/week4$ Parent exiting
./a.out & ps

```

```

[3] 428325
Child exiting
  PID TTY          TIME CMD
 17658 pts/0        00:00:00 bash
 428325 pts/0        00:00:00 a.out
 428326 pts/0        00:00:00 ps
 428327 pts/0        00:00:00 a.out
[2]  Done                  ./a.out

```

4. Write a multithreaded program that performs different sorting algorithms. The program should work as follows: the user enters on the command line the number of elements to sort and the elements themselves. The program then creates separate threads, each using a different sorting algorithm. Each thread sorts the array using its corresponding algorithm and displays the time taken to produce the result. The main thread waits for all threads to finish and then displays the final sorted array.

```
ex4.c ✕
1  #include <stdio.h>
2  #include <pthread.h>
3  #include <stdlib.h>
4  #include <time.h>
5
6  int n;
7  int arr[100];
8
9  void* bubble_sort(void* arg) {
10     clock_t start = clock();
11     int temp[100];
12     for (int i = 0; i < n; i++) temp[i] = arr[i];
13
14     for (int i = 0; i < n-1; i++)
15         for (int j = 0; j < n-i-1; j++)
16             if (temp[j] > temp[j+1]) {
17                 int t = temp[j];
18                 temp[j] = temp[j+1];
19                 temp[j+1] = t;
20             }
21
22     clock_t end = clock();
23     printf("Bubble Sort Time: %f\n", (double)(end-start)/CLOCKS_PER_SEC);
24     pthread_exit(NULL);
25 }
26
27 void* selection_sort(void* arg) {
28     clock_t start = clock();
29     int temp[100];
30     for (int i = 0; i < n; i++) temp[i] = arr[i];
31
32     for (int i = 0; i < n-1; i++) {
33         int min = i;
34         for (int j = i+1; j < n; j++)
35             if (temp[j] < temp[min]) min = j;
36         int t = temp[i];
37         temp[i] = temp[min];
38         temp[min] = t;
39     }
```

```

40
41     clock_t end = clock();
42     printf("Selection Sort Time: %f\n", (double)(end-start)/CLOCKS_PER_SEC);
43     pthread_exit(NULL);
44 }
45
46 int main() {
47     pthread_t t1, t2;
48
49     printf("Enter number of elements: ");
50     scanf("%d", &n);
51
52     printf("Enter elements:\n");
53     for (int i = 0; i < n; i++)
54         scanf("%d", &arr[i]);
55
56     pthread_create(&t1, NULL, bubble_sort, NULL);
57     pthread_create(&t2, NULL, selection_sort, NULL);
58
59     pthread_join(t1, NULL);
60     pthread_join(t2, NULL);
61
62     // Final sorted array (bubble sort)
63     for (int i = 0; i < n-1; i++)
64         for (int j = 0; j < n-i-1; j++)
65             if (arr[j] > arr[j+1]) {
66                 int t = arr[j];
67                 arr[j] = arr[j+1];
68                 arr[j+1] = t;
69             }
70
71     printf("Final Sorted Array:\n");
72     for (int i = 0; i < n; i++)
73         printf("%d ", arr[i]);
74
75     return 0;
76 }

```

```

4ITB2@debian:~/hridini_240911482/week4$ gcc ex4.c
4ITB2@debian:~/hridini_240911482/week4$ gcc Parent exiting

gcc: fatal error: no input files
compilation terminated.
[3]+  Done                  ./a.out
4ITB2@debian:~/hridini_240911482/week4$ gcc ex4.c
4ITB2@debian:~/hridini_240911482/week4$ gcc ex4.c -lpthread
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Enter number of elements: 4
Enter elements:
50
21
7
19
Bubble Sort Time: 0.000002
Selection Sort Time: 0.000003
Final Sorted Array:
7 19 21 50 4ITB2@debian:~/hridini_240911482/week4$ vi ex5.c

```

5. Write multithreaded program that generates the Fibonacci series. The program should work as follows: The user will enter on the command line the number of Fibonacci numbers that the program is to generate. The program then will create a separate thread that will generate the Fibonacci numbers, placing the sequence in data that is shared by the threads (an array is probably the most convenient data structure). When the thread finishes execution the parent will output the sequence generated by the child thread. Because the parent thread cannot begin outputting the Fibonacci sequence until the child thread finishes, this will require having the parent thread wait for the child thread to finish.

```

ex5.c ✕
1  #include <stdio.h>
2  #include <pthread.h>
3
4  int fib[100];
5  int n;
6
7  void* generate_fib(void* arg) {
8      fib[0] = 0;
9      fib[1] = 1;
10
11      for (int i = 2; i < n; i++)
12          fib[i] = fib[i-1] + fib[i-2];
13
14      pthread_exit(NULL);
15  }
16
17  int main() {
18      pthread_t thread;
19
20      printf("Enter number of Fibonacci terms: ");
21      scanf("%d", &n);
22
23      pthread_create(&thread, NULL, generate_fib, NULL);
24      pthread_join(thread, NULL);
25
26      printf("Fibonacci Series:\n");
27      for (int i = 0; i < n; i++)
28          printf("%d ", fib[i]);
29
30      return 0;
31  }

```

```

4ITB2@debian:~/hridini_240911482/week4$ gcc ex5.c
4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Enter number of Fibonacci terms: 5
Fibonacci Series:
0 1 1 2 3 4ITB2@debian:~/hridini_240911482/week4$ ./a.out
Enter number of Fibonacci terms: 19
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4ITB2@debian:~/hridini_24091

```


LAB NO:5

CPU SCHEDULING ALGORITHMS

Lab Exercise

1. Consider the following set of processes, with length of the CPU burst given in milliseconds:

<i>Process</i>	<i>Arrival time</i>	<i>Burst time</i>	<i>Priority</i>
P1	0	60	3
P2	3	30	2
P3	4	40	1
P4	9	10	4

Write a menu driven C/C++ program to simulate the following CPU scheduling algorithms.

Display Gantt chart showing the order of execution of each process.

Compute waiting time and

turnaround time for each process. Hence, compute average waiting time and average turnaround time.

(i) FCFS (ii) SRTF (iii) Round-Robin (quantum = 10) iv) non-preemptive priority (higher the number higher the priority)

```
1  #include <stdio.h>
2
3  #define MAX 20
4
5  void fcfs(int n, int at[], int bt[]);
6  void srtf(int n, int at[], int bt[]);
7  void roundRobin(int n, int at[], int bt[], int quantum);
8  void priorityNP(int n, int at[], int bt[], int pr[]);
9
10 void fcfs(int n, int at[], int bt[]) {
11     int ct[MAX], wt[MAX], tat[MAX];
12     int time = 0;
13
14     printf("\nGantt Chart:\n");
15
16     for(int i = 0; i < n; i++) {
17
18         if(time < at[i])
19             time = at[i];
20
21         time += bt[i];
22         ct[i] = time;
23
24         tat[i] = ct[i] - at[i];
25         wt[i] = tat[i] - bt[i];
26
27         printf("P%d ", i+1);
28     }
29
30     float avg_wt = 0, avg_tat = 0;
31
32     printf("\n\nProcess\tWT\tTAT\n");
33     for(int i = 0; i < n; i++) {
34         printf("P%d\t%d\t%d\n", i+1, wt[i], tat[i]);
35         avg_wt += wt[i];
36         avg_tat += tat[i];
37     }
38
39     printf("Average WT = %.2f\n", avg_wt/n);
40     printf("Average TAT = %.2f\n", avg_tat/n);
41 }
```

```

void srtf(int n, int at[], int bt[]) {
    int rt[MAX], ct[MAX], wt[MAX], tat[MAX];
    int completed = 0, time = 0, min, shortest;

    for(int i = 0; i < n; i++)
        rt[i] = bt[i];
    printf("\nGantt Chart:\n");
    while(completed != n) {

        min = 9999;
        shortest = -1;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] > 0 && rt[i] < min) {
                min = rt[i];
                shortest = i;
            }
        }
        if(shortest == -1) {
            time++;
            continue;
        }

        rt[shortest]--;
        printf("P%d ", shortest+1);
        time++;

        if(rt[shortest] == 0) {
            completed++;
            ct[shortest] = time;
        }
    }
    float avg_wt = 0, avg_tat = 0;
    printf("\n\nProcess\tWT\tTAT\n");
    for(int i = 0; i < n; i++) {
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        printf("P%d\t%d\t%d\n", i+1, wt[i], tat[i]);

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    printf("Average WT = %.2f\n", avg_wt/n);
    printf("Average TAT = %.2f\n", avg_tat/n);
}

```

```

void roundRobin(int n, int at[], int bt[], int quantum) {
    int rt[MAX], wt[MAX], tat[MAX], ct[MAX];
    int time = 0, completed = 0;

    for(int i = 0; i < n; i++)
        rt[i] = bt[i];

    printf("\nGantt Chart:\n");

    while(completed != n) {

        for(int i = 0; i < n; i++) {

            if(rt[i] > 0 && at[i] <= time) {

                if(rt[i] > quantum) {
                    time += quantum;
                    rt[i] -= quantum;
                    printf("P%d ", i+1);
                }
                else {
                    time += rt[i];
                    printf("P%d ", i+1);
                    rt[i] = 0;
                    ct[i] = time;
                    completed++;
                }
            }
        }

        float avg_wt = 0, avg_tat = 0;
        printf("\n\nProcess\tWT\tTAT\n");

        for(int i = 0; i < n; i++) {
            tat[i] = ct[i] - at[i];
            wt[i] = tat[i] - bt[i];

            printf("P%d\t%d\t%d\n", i+1, wt[i], tat[i]);

            avg_wt += wt[i];
            avg_tat += tat[i];
        }
        printf("Average WT = %.2f\n", avg_wt/n);
        printf("Average TAT = %.2f\n", avg_tat/n);
    }
}

```

```

void priorityNP(int n, int at[], int bt[], int pr[]) {
    int completed = 0, time = 0;
    int visited[MAX] = {0};
    int ct[MAX], wt[MAX], tat[MAX];

    printf("\nGantt Chart:\n");

    while(completed != n) {

        int idx = -1;
        int highest = -1;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && visited[i] == 0) {
                if(pr[i] > highest) {
                    highest = pr[i];
                    idx = i;
                }
            }

            if(idx == -1) {
                time++;
                continue;
            }

            time += bt[idx];
            ct[idx] = time;
            visited[idx] = 1;
            completed++;

            printf("P%d ", idx+1);
        }
        float avg_wt = 0, avg_tat = 0;
        printf("\n\nProcess\tWT\tTAT\n");

        for(int i = 0; i < n; i++) {
            tat[i] = ct[i] - at[i];
            wt[i] = tat[i] - bt[i];

            printf("P%d\t%d\t%d\n", i+1, wt[i], tat[i]);

            avg_wt += wt[i];
            avg_tat += tat[i];
        }
        printf("Average WT = %.2f\n", avg_wt/n);
        printf("Average TAT = %.2f\n", avg_tat/n);
    }
}

```

```

int main() {
    int n, choice;
    int at[MAX], bt[MAX], pr[MAX];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++) {
        printf("Process %d\n", i+1);
        printf("Arrival Time: ");
        scanf("%d", &at[i]);
        printf("Burst Time: ");
        scanf("%d", &bt[i]);
        printf("Priority: ");
        scanf("%d", &pr[i]);
    }

    do {
        printf("\n1.FCFs\n2.SRTF\n3.Round Robin\n4.Pri
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1: fcfs(n, at, bt); break;
            case 2: srtf(n, at, bt); break;
            case 3: roundRobin(n, at, bt, 10); break;
            case 4: priorityNP(n, at, bt, pr); break;
        }
    } while(choice != 5);

    return 0;
}

```

```
4ITB2@debian:~/hridini_240911482/lab5$ gcc Q1.c
```

```
4ITB2@debian:~/hridini_240911482/lab5$ ./a.out
```

```
Enter number of processes: 4
```

```
Process 1
```

```
Arrival Time: 0
```

```
Burst Time: 60
```

```
Priority: 3
```

```
Process 2
```

```
Arrival Time: 3
```

```
Burst Time: 30
```

```
Priority: 2
```

```
Process 3
```

```
Arrival Time: 4
```

```
Burst Time: 40
```

```
Priority: 1
```

```
Process 4
```

```
Arrival Time: 9
```

```
Burst Time: 10
```

```
Priority: 4
```

```
1.FCFS
```

```
2.SRTF
```

```
3.Round Robin
```

```
4.Priority (Non-preemptive)
```

```
5.Exit
```

```
Enter choice: 1
```

```
Gantt Chart:
```

```
P1 P2 P3 P4
```

```
Process WT      TAT
```

```
P1      0       60
```

```
P2     57       87
```

```
P3     86      126
```

```
P4    121      131
```

```
Average WT = 66.00
```

```
Average TAT = 101.00
```

```
1.FCFS
```

```
2.SRTF
```

```
3.Round Robin
```

```
4.Priority (Non-preemptive)
```

```
5.Exit
```

Enter choice: 2

Gantt Chart:

P1 P1 P1 P2 P2 P2 P2 P2 P2 P4 P4 P4 P4 P4 P4 P4 P4 P2 P2 P2 P2 P2 P2 P2 P2 P2 P
2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P3 P3 P3 P3 P3 P3 P3 P3 P3 P3 P3 P3
P3 P1 P1
P1 P
1 P1

Process	WT	TAT
P1	80	140
P2	10	40
P3	39	79
P4	0	10

Average WT = 32.25
Average TAT = 67.25

- 1.FCFS
- 2.SRTF
- 3.Round Robin
- 4.Priority (Non-preemptive)
- 5.Exit

Enter choice: 3

Gantt Chart:

P1 P2 P3 P4 P1 P2 P3 P1 P2 P3 P1 P3 P1 P1

Process	WT	TAT
P1	80	140
P2	57	87
P3	76	116
P4	21	31

Average WT = 58.50
Average TAT = 93.50

- 1.FCFS
- 2.SRTF
- 3.Round Robin
- 4.Priority (Non-preemptive)
- 5.Exit


```
Enter choice: 4

Gantt Chart:
P1 P4 P2 P3

Process WT      TAT
P1      0       60
P2      67       97
P3      96      136
P4      51       61
Average WT = 53.50
Average TAT = 88.50

1.FCFS
2.SRTF
3.Round Robin
4.Priority (Non-preemptive)
5.Exit
Enter choice: 5
```

2. Write a C program to simulate multi-level feedback queue scheduling algorithm.

Q2.c ✕

```
1  #include <stdio.h>
2  #define MAX 20
3  int main() {
4      int n;
5      int at[MAX], bt[MAX], rt[MAX];
6      int ct[MAX], wt[MAX], tat[MAX];
7      int time = 0;
8
9      printf("Enter number of processes: ");
10     scanf("%d", &n);
11
12     for(int i = 0; i < n; i++) {
13         printf("\nProcess %d\n", i+1);
14         printf("Arrival Time: ");
15         scanf("%d", &at[i]);
16         printf("Burst Time: ");
17         scanf("%d", &bt[i]);
18         rt[i] = bt[i]; // copy burst time
19     }
20     printf("\nGantt Chart:\n");
21     // ----- Q0 (Quantum = 8) -----
22     for(int i = 0; i < n; i++) {
23         if(rt[i] > 0) {
24             if(rt[i] > 8) {
25                 time += 8;
26                 rt[i] -= 8;
27                 printf("P%d ", i+1);
28             } else {
29                 time += rt[i];
30                 printf("P%d ", i+1);
31                 rt[i] = 0;
32                 ct[i] = time;
33             }
34         }
35     }
```

```

34
35 // ----- Q1 (Quantum = 16) -----
36 for(int i = 0; i < n; i++) {
37     if(rt[i] > 0) {
38         if(rt[i] > 16) {
39             time += 16;
40             rt[i] -= 16;
41             printf("P%d ", i+1);
42         } else {
43             time += rt[i];
44             printf("P%d ", i+1);
45             rt[i] = 0;
46             ct[i] = time;
47         }
48     }
49
50 // ----- Q2 (FCFS) -----
51 for(int i = 0; i < n; i++) {
52     if(rt[i] > 0) {
53         time += rt[i];
54         printf("P%d ", i+1);
55         rt[i] = 0;
56         ct[i] = time;
57     }
58 }
59 float avg_wt = 0, avg_tat = 0;
60 printf("\n\nProcess\tWT\tTAT\n");
61 for(int i = 0; i < n; i++) {
62     tat[i] = ct[i] - at[i];
63     wt[i] = tat[i] - bt[i];
64     printf("P%d\t%d\t%d\n", i+1, wt[i], tat[i]);
65     avg_wt += wt[i];
66     avg_tat += tat[i];
67 }
68
69 printf("\nAverage Waiting Time = %.2f", avg_wt/n);
70 printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);
71
72 return 0;

```

```
4ITB2@debian:~/hridini_240911482/lab5$ vi Q2.c
4ITB2@debian:~/hridini_240911482/lab5$ gcc Q2.c
/usr/bin/ld: /usr/lib/gcc/x86_64-linux-gnu/12/../../../../x86_64-linux-gnu/Scrt1.o: in function `_start':
(.text+0x17): undefined reference to `main'
collect2: error: ld returned 1 exit status
4ITB2@debian:~/hridini_240911482/lab5$ vi Q2.c
4ITB2@debian:~/hridini_240911482/lab5$ vi Q2.c
4ITB2@debian:~/hridini_240911482/lab5$ gcc Q2.c
4ITB2@debian:~/hridini_240911482/lab5$ ./a.out
Enter number of processes: 2

Process 1
Arrival Time: 0
Burst Time: 3

Process 2
Arrival Time: 1
Burst Time: 2

Gantt Chart:
P1 P2

Process WT      TAT
P1      0       3
P2      2       4

Average Waiting Time = 1.00
Average Turnaround Time = 3.50
```

3. Write a C program to simulate Earliest-Deadline-First scheduling for realtime systems.

```
Q3.c x
1  #include <stdio.h>
2  #define MAX 20
3
4  int main() {
5      int n;
6      int at[MAX], bt[MAX], deadline[MAX];
7      int rt[MAX], ct[MAX], wt[MAX], tat[MAX];
8      int time = 0, completed = 0;
9      printf("Enter number of processes: ");
10     scanf("%d", &n);
11     for(int i = 0; i < n; i++) {
12         printf("\nProcess %d\n", i+1);
13         printf("Arrival Time: ");
14         scanf("%d", &at[i]);
15         printf("Burst Time: ");
16         scanf("%d", &bt[i]);
17         printf("Deadline: ");
18         scanf("%d", &deadline[i]);
19
20         rt[i] = bt[i];
21     }
22     printf("\nGantt Chart:\n");
23     while(completed != n) {
24         int idx = -1;
25         int earliest = 9999;
26
27         for(int i = 0; i < n; i++) {
28             if(at[i] <= time && rt[i] > 0) {
29                 if(deadline[i] < earliest) {
30                     earliest = deadline[i];
31                     idx = i;
32                 }
33             }
34         }
35         if(idx == -1) {
36             time++;
37             continue;
38         }
39         rt[idx]--;
40         printf("P%d ", idx+1);
41         time++;
42         if(rt[idx] == 0) {
43             ct[idx] = time;
44             completed++;
45         }
46     }
47 }
```

```

5
6     float avg_wt = 0, avg_tat = 0;
7     printf("\n\nProcess\tWT\tTAT\n");
8     for(int i = 0; i < n; i++) {
9         tat[i] = ct[i] - at[i];
10        wt[i] = tat[i] - bt[i];
11        printf("P%d\t%d\t%d\n", i+1, wt[i], tat[i]);
12        avg_wt += wt[i];
13        avg_tat += tat[i];
14
15    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
16    printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);
17
18    return 0;
19 }

```

```

4ITB2@debian:~/hridini_240911482/lab5$ vi Q3.c
4ITB2@debian:~/hridini_240911482/lab5$ gcc Q3.c
4ITB2@debian:~/hridini_240911482/lab5$ ./a.out
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 60
Deadline:

^[
Process 2
Arrival Time: Burst Time: Deadline:
Process 3
Arrival Time: Burst Time: Deadline:
Process 4
Arrival Time: Burst Time: Deadline:
Gantt Chart:

^[^[^[^[
^C
4ITB2@debian:~/hridini_240911482/lab5$ ./a.out
Enter number of processes: 2

Process 1
Arrival Time: 50
Burst Time: 25
Deadline: 50

Process 2
Arrival Time: 80
Burst Time: 35
Deadline: 80

Gantt Chart:
P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P1 P2 P2 P2 P
2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2 P2
P2 P2 P2

Process WT      TAT
P1      0       25
P2      0       35

Average Waiting Time = 0.00
Average Turnaround Time = 30.00

```

LAB NO.: 6

INTERPROCESS COMMUNICATION

Lab Exercises:

1. Process A wants to send a number to Process B. Once received, Process B has to check whether the number is palindrome or not. Write a C program to implement this interprocess communication using message queue.

```
sender.c x receiver.c x
1 // sender.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <sys/ipc.h>
5 #include <sys/msg.h>
6 #include <string.h>
7
8 struct msg_st {
9     long int msg_type;
10    char text[100];
11 };
12 int main() {
13     int msgid;
14     struct msg_st data;
15     msgid = msgget((key_t)1234, 0666 | IPC_CREAT);
16     printf("Enter a number: ");
17     scanf("%s", data.text);
18     data.msg_type = 1;
19     msgsnd(msgid, (void *)&data, sizeof(data.text), 0);
20     printf("Number sent to Process B\n");
21     return 0;
22 }
```



```
sender.c x receiver.c x
1 // receiver.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <sys/ipc.h>
5 #include <sys/msg.h>
6 #include <string.h>
7 struct msg_st {
8     long int msg_type;
9     char text[100];
10 };
11 int isPalindrome(char num[]) {
12     int i, len = strlen(num);
13     for(i = 0; i < len/2; i++) {
14         if(num[i] != num[len - i - 1])
15             return 0;
16     }
17     return 1;
18 }
19 int main() {
20     int msgid;
21     struct msg_st data;
22     msgid = msgget((key_t)1234, 0666 | IPC_CREAT);
23     msgrcv(msgid, (void *)&data, sizeof(data.text), 0, 0);
24     printf("Received number: %s\n", data.text);
25     if(isPalindrome(data.text))
26         printf("It is Palindrome\n");
27     else
28         printf("Not a Palindrome\n");
29     msgctl(msgid, IPC_RMID, 0);
30     return 0;
}
```

2. Write a producer and consumer program in C using FIFO queue. The producer should write a set of 4 integers into the FIFO queue and the consumer should display the 4 integers.

producer.c ✕

consumer.c ✕

```
1 // producer_fifo.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <unistd.h>
5 #include <fcntl.h>
6 #include <sys/stat.h>
7
8 #define FIFO_NAME "myfifo"
9
10 int main() {
11     int fd;
12     int numbers[4];
13
14     mkfifo(FIFO_NAME, 0666);
15
16     fd = open(FIFO_NAME, O_WRONLY);
17
18     printf("Enter 4 integers:\n");
19     for(int i=0;i<4;i++)
20         scanf("%d",&numbers[i]);
21
22     write(fd, numbers, sizeof(numbers));
23
24     close(fd);
25     return 0;
26 }
```

```
producer.c x consumer.c x
1 // consumer_fifo.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <unistd.h>
5 #include <fcntl.h>
6
7 #define FIFO_NAME "myfifo"
8
9 int main() {
10     int fd;
11     int numbers[4];
12
13     fd = open(FIFO_NAME, O_RDONLY);
14
15     read(fd, numbers, sizeof(numbers));
16
17     printf("Received numbers:\n");
18     for(int i=0;i<4;i++)
19         printf("%d ", numbers[i]);
20
21     printf("\n");
22
23     close(fd);
24     unlink(FIFO_NAME);
25
26     return 0;}
```

3. Implement a parent process, which sends an English alphabet to child process using shared memory. Child process responds back with next English alphabet to the parent. Parent displays the reply from the Child.

```
parent_child_alp_shm.c X
1 // parent_child_shm.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <sys/ipc.h>
5 #include <sys/shm.h>
6 #include <unistd.h>
7 #include <sys/wait.h>
8
9 int main() {
10     int shmid;
11     char *shared_mem;
12
13     shmid = shmget(1234, 10, 0666 | IPC_CREAT);
14     shared_mem = (char*) shmat(shmid, NULL, 0);
15     printf("Enter an alphabet: ");
16     scanf(" %c", shared_mem);
17     if(fork() == 0) {
18         // Child
19         *shared_mem = *shared_mem + 1;
20         exit(0);}
21     else {
22         wait(NULL);
23         printf("Next alphabet from child: %c\n", *shared_mem);}
24     shmdt(shared_mem);
25     shmctl(shmid, IPC_RMID, NULL);
26     return 0;
27 }
```

4. Write a producer-consumer program in C in which producer writes a set of words into shared memory and then consumer reads the set of words from the shared memory. The shared memory need to be detached and deleted after use.

consumer_shm.c	producer_shm.c
<pre> 1 // consumer_shm.c 2 #include <stdio.h> 3 #include <stdlib.h> 4 #include <sys/ipc.h> 5 #include <sys/shm.h> 6 #include "shm_words.h" 7 8 int main() { 9 int shmid; 10 struct shared_data *shared; 11 12 shmid = shmget(1234, sizeof(struct shared_data), 0666 IPC_CREAT); 13 shared = shmat(shmid, NULL, 0); 14 while(shared->written != 1); 15 printf("Received word: %s\n", shared->text); 16 shmdt(shared); 17 shmctl(shmid, IPC_RMID, NULL); 18 return 0; 19 }</pre>	

consumer_shm.c	producer_shm.c
	<pre> 1 // producer_shm.c 2 #include <stdio.h> 3 #include <stdlib.h> 4 #include <string.h> 5 #include <sys/ipc.h> 6 #include <sys/shm.h> 7 #include "shm_words.h" 8 9 int main() { 10 int shmid; 11 struct shared_data *shared; 12 13 shmid = shmget(1234, sizeof(struct shared_data), 0666 IPC_CREAT); 14 shared = shmat(shmid, NULL, 0); 15 printf("Enter a word: "); 16 scanf("%s", shared->text); 17 shared->written = 1; 18 shmdt(shared); 19 return 0; 20 }</pre>

```
4ITB2@debian: ~/hridini_240911482/week6
File Edit View Search Terminal Tabs Help
4ITB2@debian: ~/hridini_240911482/week6 x 4ITB2@debian: ~/hridini_240911482/week6 x + ▾

4ITB2@debian:~/hridini_240911482/week6$ ./receiver
Received number: 69
Not a Palindrome
4ITB2@debian:~/hridini_240911482/week6$ ./receiver
Received number: 9669
It is Palindrome
4ITB2@debian:~/hridini_240911482/week6$ vi producer.c
4ITB2@debian:~/hridini_240911482/week6$ vi consumer.c
4ITB2@debian:~/hridini_240911482/week6$ gcc producer.c -o producer
4ITB2@debian:~/hridini_240911482/week6$ ./producer
Enter 4 integers:
50
72
19
420
4ITB2@debian:~/hridini_240911482/week6$ vi receiver.c
4ITB2@debian:~/hridini_240911482/week6$ vi parent_child_alp_shm.c
4ITB2@debian:~/hridini_240911482/week6$ gcc parent_child_alp_shm.c -o run
parent_child_alp_shm.c: In function 'main':
parent_child_alp_shm.c:24:9: warning: implicit declaration of function 'wait' [-Wimplicit-function-declaration]
   24 |         wait(NULL);
      |         ^~~~~
4ITB2@debian:~/hridini_240911482/week6$ vi parent_child_alp_shm.c
4ITB2@debian:~/hridini_240911482/week6$ gcc parent_child_alp_shm.c -o run
4ITB2@debian:~/hridini_240911482/week6$ ./run
Enter an alphabet: a
Next alphabet from child: b
4ITB2@debian:~/hridini_240911482/week6$ ./run
Enter an alphabet: z
Next alphabet from child: {
4ITB2@debian:~/hridini_240911482/week6$ ./run
Enter an alphabet: w
Next alphabet from child: x
4ITB2@debian:~/hridini_240911482/week6$ ./run
Enter an alphabet: Q
Next alphabet from child: R
4ITB2@debian:~/hridini_240911482/week6$ vi sender2.c
4ITB2@debian:~/hridini_240911482/week6$ vi shm_words.h
4ITB2@debian:~/hridini_240911482/week6$ vi producer_shm.c
4ITB2@debian:~/hridini_240911482/week6$ vi consumer_shm.c
4ITB2@debian:~/hridini_240911482/week6$ gcc producer_shm.c -o producer_shm
4ITB2@debian:~/hridini_240911482/week6$ ./producer_shm
Enter a word: hridini
```

```
4ITB2@debian:~/hridini_240911482/week6$ ^[[200~gcc sender.c -o sender
bash: $'\E[200~gcc': command not found
4ITB2@debian:~/hridini_240911482/week6$ ~gcc sender.c -o sender
bash: ~gcc: command not found
4ITB2@debian:~/hridini_240911482/week6$ gcc sender.c -o sender
4ITB2@debian:~/hridini_240911482/week6$ ./sender
Enter a number: 69
Number sent to Process B
4ITB2@debian:~/hridini_240911482/week6$ ./sender
Enter a number: 9669
Number sent to Process B
4ITB2@debian:~/hridini_240911482/week6$ gcc consumer.c -o consumer
4ITB2@debian:~/hridini_240911482/week6$ ./consumer
Received numbers:
0 0 792493936 32528
4ITB2@debian:~/hridini_240911482/week6$ ./consumer
Received numbers:
50 72 19 420
4ITB2@debian:~/hridini_240911482/week6$ gcc consumer_shm.c -o consumer_shm.c
gcc: fatal error: input file 'consumer_shm.c' is the same as output file
compilation terminated.
4ITB2@debian:~/hridini_240911482/week6$ gcc consumer_shm.c -o consumer_shm
4ITB2@debian:~/hridini_240911482/week6$ ./consumer_shm
Received word: hridini
4ITB2@debian:~/hridini_240911482/week6$
```

LAB NO: 7

CLASSICAL PROBLEMS OF SYNCHRONIZATION

Lab Exercise

1. Modify the above Producer-Consumer program so that, a producer can produce at the most 10 items more than what the consumer has consumed.

```
4ITB2@debian:~/hridini_240911482/lab7$ gcc Q1.c
4ITB2@debian:~/hridini_240911482/lab7$ ./a.out
Produced: 0 | Total Produced: 1
Consumed: 0 | Total Consumed: 1
Produced: 1 | Total Produced: 2
Consumed: 1 | Total Consumed: 2
Produced: 2 | Total Produced: 3
Consumed: 2 | Total Consumed: 3
Produced: 3 | Total Produced: 4
Consumed: 3 | Total Consumed: 4
Produced: 4 | Total Produced: 5
Consumed: 4 | Total Consumed: 5
Produced: 5 | Total Produced: 6
Consumed: 5 | Total Consumed: 6
Produced: 6 | Total Produced: 7
Consumed: 6 | Total Consumed: 7
Produced: 7 | Total Produced: 8
Consumed: 7 | Total Consumed: 8
Produced: 8 | Total Produced: 9
Consumed: 8 | Total Consumed: 9
Produced: 9 | Total Produced: 10
Consumed: 9 | Total Consumed: 10
Produced: 10 | Total Produced: 11
Consumed: 10 | Total Consumed: 11
^C
```


2. Write a C/C++ program for first readers-writers problem using semaphores.

```
4ITB2@debian:~/hridini_240911482/lab7$ gcc Q2.c
4ITB2@debian:~/hridini_240911482/lab7$ ./a.out
Reader 1 is reading data = 0
Reader 2 is reading data = 0
Reader 3 is reading data = 0
Writer 1 is writing data = 10
Writer 2 is writing data = 20
```

Q1.c ✕

Q2.c ✕

```
1  #include<stdio.h>
2  #include<pthread.h>
3  #include<semaphore.h>
4  #include<unistd.h>
5
6  #define SIZE 5
7  #define MAX_ITEMS 20
8
9  int buf[SIZE];
10 int f = -1, r = -1;
11 int produced = 0;
12 int consumed = 0;
13 sem_t mutex, full, empty, limit;
14 void* produce(void* arg){
15     for(int i = 0; i < MAX_ITEMS; i++) {
16         sem_wait(&limit);          // ensure producer doesn't exceed 10 extra items
17         sem_wait(&empty);
18         sem_wait(&mutex);
19
20         r = (r + 1) % SIZE;
21         buf[r] = i;
22         produced++;
23         printf("Produced: %d | Total Produced: %d\n", i, produced);
24         sem_post(&mutex);
25         sem_post(&full);
26         sleep(1);}
27     return NULL;}
28
29 void* consume(void* arg){
30     for(int i = 0; i < MAX_ITEMS; i++){
31         sem_wait(&full);
32         sem_wait(&mutex);
33
34         f = (f + 1) % SIZE;
35         int item = buf[f];
36         consumed++;
37         printf("Consumed: %d | Total Consumed: %d\n", item, consumed);
38         sem_post(&mutex);
39         sem_post(&empty);
40         sem_post(&limit);          // allow producer to produce more
41         sleep(1);}
42     return NULL;}
```

```
44  int main(){
45      pthread_t t1, t2;
46      sem_init(&mutex, 0, 1);
47      sem_init(&full, 0, 0);
48      sem_init(&empty, 0, SIZE);
49      sem_init(&limit, 0, 10); // producer can be ahead by at most 10
50      pthread_create(&t1, NULL, produce, NULL);
51      pthread_create(&t2, NULL, consume, NULL);
52      pthread_join(t1, NULL);
53      pthread_join(t2, NULL);
54      sem_destroy(&mutex);
55      sem_destroy(&full);
56      sem_destroy(&empty);
57      sem_destroy(&limit);
58      return 0;}
```

Q1.c X

Q2.c X

```
1  #include<stdio.h>
2  #include<pthread.h>
3  #include<semaphore.h>
4  #include<unistd.h>
5
6  sem_t mutex, wrt;
7  int readcount = 0;
8  int data = 0;
9
10 void* reader(void* arg){
11     int id = *(int*)arg;
12     sem_wait(&mutex);
13     readcount++;
14     if(readcount == 1)
15         sem_wait(&wrt);    // first reader blocks writers
16     sem_post(&mutex);
17     printf("Reader %d is reading data = %d\n", id, data);
18     sleep(1);
19     sem_wait(&mutex);
20     readcount--;
21     if(readcount == 0)
22         sem_post(&wrt);    // last reader allows writers
23     sem_post(&mutex);
24     return NULL;}
25
26 void* writer(void* arg){
27     int id = *(int*)arg;
28     sem_wait(&wrt);
29     data += 10;
30     printf("Writer %d is writing data = %d\n", id, data);
31     sleep(1);
32     sem_post(&wrt);
33     return NULL;
34 }
```

```
36  int main(){
37      pthread_t r[3], w[2];
38      int r_id[3] = {1,2,3};
39      int w_id[2] = {1,2};
40
41      sem_init(&mutex, 0, 1);
42      sem_init(&wrt, 0, 1);
43
44      for(int i = 0; i < 3; i++)
45          pthread_create(&r[i], NULL, reader, &r_id[i]);
46
47      for(int i = 0; i < 2; i++)
48          pthread_create(&w[i], NULL, writer, &w_id[i]);
49
50      for(int i = 0; i < 3; i++)
51          pthread_join(r[i], NULL);
52
53      for(int i = 0; i < 2; i++)
54          pthread_join(w[i], NULL);
55
56      sem_destroy(&mutex);
57      sem_destroy(&wrt);
58      return 0;
59  }
```